MUHAMMAD ASHIR

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RESEARCH PROFILE

Award-winning MS researcher building efficient, distributed vision-language systems. My work applying and extending Mixture-of-Experts (MoE) to edge inference won Best Presentation at ICCAI 2026 (ACM) and was the first of its kind deployed across a distributed single-board-computer cluster.

My research interests span LLMs, efficient machine learning, deep learning, distributed & edge AI (on-device inference, multi-node deployment), multimodal AI, and trustworthy & reliable AI. I work at the intersection of these areas to build real-world deployable AI systems.

EDUCATION

M.S. in Software and Information Science (expected) Apr 2025 – Mar 2027

Iwate Prefectural University, Graduate School of Software and Information Science — Iwate, Japan

Thesis: Optimization and Implementation of Vision-Language Models on Single-Board Computer Clusters

Advisors: Prof. Hirokazu Madokoro & Prof. Stephanie Nix. Funding: MEXT (Japanese Government) Scholarship — full tuition and stipend.

CGPA: 3.71/4.0 (converted from official transcript per Scholaro's Japan 4-scale-university grading table: Very Good → A, Good → B; credit-weighted).

B.S. in Information Technology Sep 2017 – Sep 2021

Government College University Faisalabad — Faisalabad, Pakistan

CGPA: 3.20/4.00.

Final-year project: Human Computer Interaction on a Food Menu Application and Website. Relevant coursework: Artificial Intelligence, Discrete Mathematics, Linear Algebra, Calculus, Probability & Statistics, Cloud Computing, Computer Networks, Database Systems, Software Engineering, Object-Oriented Analysis & Design, Web Development.

PUBLICATIONS

- [C2] **Muhammad Ashir**, H. Madokoro, and S. Nix. “Mixture of Tiny Experts for Resource-Constrained Vision–Language Models on Single-Board Computer Clusters.” Proceedings of the 12th International Conference on Computing and Artificial Intelligence (ICCAI 2026), Okinawa, Japan, Apr. 2026. Published by ACM (Scopus, EI Compendex indexed). — Best Presentation Award; rated “Highly Innovative” by peer reviewers.
- [C1] **A. Muhammad**, H. Madokoro, and S. Nix. “Optimization and Implementation of Qwen-VL on a Single-Board Cluster Computer.” Proceedings of SI2025 - 26th Annual Conference of SICE Systems Integration Division, Hiroshima, Japan, pp. 203–204, Dec. 2025. - Outstanding Presentation Award.
- [J1 — In Preparation] **Muhammad Ashir**, H. Madokoro, and S. Nix. “Lightweight Edge AI for Vision-Language Models: A Mixture of Tiny Experts Approach for Distributed Inference on Single Board Computer Clusters.” Target venue: IEEE Access. Target submission: November 2026.

RESEARCH EXPERIENCE

Graduate Researcher — Madokoro & Nix Laboratory Sep 2024 – Present

Iwate Prefectural University, Morioka, Japan

- Designed and implemented MoTE-VLM - a distributed Mixture-of-Experts framework deploying Qwen2.5-VL-3B-Instruct across a 3-node Raspberry Pi 5 cluster using TinyExpert modules with top-1 routing.
- Demonstrated feasibility of billion-parameter vision language model inference on commodity edge hardware, achieving 48.55 ms latency and 1.80× speedup with only 5.20 ms routing overhead.
- Conducted a routing-stability verification study on COCO data to validate expert-selection consistency under distributional shift.
- Independently led the full research pipeline literature review, system design, implementation, experimental evaluation, and manuscript preparation in LaTeX (ACM sigconf and IEEEtran), including proposing the MoE routing approach after initial dense-model deployment proved infeasible on Pi hardware.

HONOURS AND AWARDS

- Best Presentation Award, ICCAI 2026, 12th International Conference on Computing and Artificial Intelligence, Okinawa, Japan, Apr. 2026.
- Outstanding Research Presentation Award, SI2025 Hiroshima, SICE Systems Integration Division, Feb. 2026.
- “Highly Innovative” Rating, ICCAI 2026, awarded for the conference paper during peer review.
- MEXT Japanese Government Scholarship, competitive, fully funded master's scholarship awarded by the Government of Japan, 2025.

TECHNICAL SKILLS

Programming: Python, C++

ML / Deep Learning: PyTorch, Hugging Face Transformers, Vision-Language Models, Large Language Models, Mixture-of-Experts architectures, model quantization and compression, multimodal AI, agentic AI systems

Mathematics: Discrete Mathematics, Linear Algebra, Calculus, Probability & Statistics

Systems & Hardware: Distributed inference on Raspberry Pi clusters, Linux system administration, GPU/edge deployment, single-board computer clusters

Tools & Workflow: Git/GitHub, Linux shell, LaTeX (IEEEtran & ACM sigconf), Bash scripting

Research Skills: Experimental design, performance profiling and benchmarking, scientific writing, peer-reviewed publication, conference presentation

LANGUAGES

English: Independent user (CEFR B2); full professional working proficiency for research and academic writing.

Japanese: Basic user (CEFR A2) for daily and academic-environment conversation.

Urdu: Native speaker.

REFERENCES

Prof. Hirokazu Madokoro, Faculty of Software and Information Science, Iwate Prefectural University: hirokazu_m@iwate-pu.ac.jp

Prof. Stephanie Nix, Faculty of Software and Information Science, Iwate Prefectural University: nix_s@iwate-pu.ac.jp